

L5: Managing Tables

RECAP from Lab #4

- Update / Delete Data

```
UPDATE
employees
SET
  lastname = 'Hill',
  email = 'mary.hill@classicmodelcars.com'
WHERE
  employeeNumber = 1056;
```

```
DELETE T1
FROM T1 LEFT JOIN
      T2 ON T1.key = T2.key
WHERE
      T2.key IS NULL;
```

Managing Table Topics

- Create Tables (Self-study in Lect.1)
- Drop Tables (Self-study in Lect.1)
- Add Columns (Lect.2)
- Drop Columns (Lect.2)
- Rename Tables
- AUTO_INCREMENT
- Temporary Tables
- Generated Columns
- Copy Table

5.1 Recap: CREATE TABLE

Syntax

```
CREATE TABLE [IF NOT EXISTS] table_name (  
    column1 datatype constraints,  
    column2 datatype constraints,  
    ...  
);
```

Example

```
CREATE TABLE tasks (  
    id INT PRIMARY KEY,  
    title VARCHAR(255) NOT NULL,  
    start_date DATE,  
    due_date DATE  
);
```

5.1 Recap: DROP TABLE

Syntax

```
DROP [TEMPORARY] TABLE [IF EXISTS] table_name ;
```

Example

```
DROP TABLE insurances;
```

5.1 Recap: View Databases & Tables

Syntax

```
SHOW DATABASES;
```

```
SHOW TABLES;
```

5.1 Recap: Add – Drop Column

Syntax

```
ALTER TABLE table_name  
ADD COLUMN new_column_name data_type  
[FIRST | AFTER existing_column];
```

Syntax

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

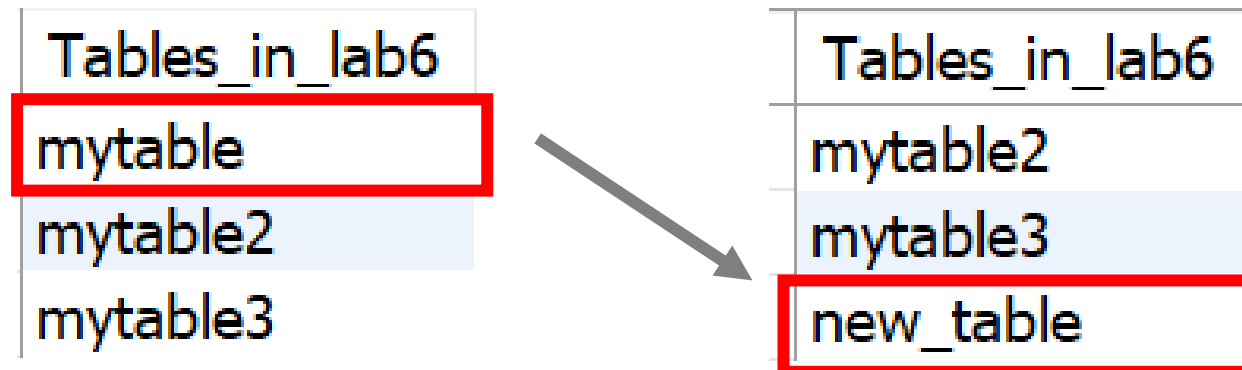
5.2 RENAME TABLE Statement

Syntax

```
RENAME TABLE table_name  
TO new_table_name;
```

Example

```
RENAME TABLE myTable  
TO new_table;
```



5.3 AUTO_INCREMENT attribute

The `AUTO_INCREMENT` attribute to automatically generate unique integer values for a column whenever you insert a new row into the table.

Example

```
CREATE TABLE contacts(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL,  
    email VARCHAR(320) NOT NULL  
);
```


AUTO_INCREMENT attribute

```
INSERT INTO contacts(name, email)
VALUES('John Doe', 'john.doe@mysqлтutorial.org');
```

```
INSERT INTO contacts(name, email)
VALUES('Jane Doe', 'jane.doe@mysqлтutorial.org');
```

```
INSERT INTO contacts(name, email)
VALUES('Bob Climo', 'bob.climo@mysqлтutorial.org');
```

id	name	email
NULL	NULL	NULL

id	name	email
1	John Doe	john.doe@mysqлтutorial.org

id	name	email
1	John Doe	john.doe@mysqлтutorial.org
2	Jane Doe	jane.doe@mysqлтutorial.org

id	name	email
1	John Doe	john.doe@mysqлтutorial.org
2	Jane Doe	jane.doe@mysqлтutorial.org
3	Bob Climo	bob.climo@mysqлтutorial.org

5.4 Temporary Tables

A temporary table is a special type of table that allows you to store a temporary result set, which you can reuse several times in a single session.

A temporary table is handy when it is impossible or expensive to query data that requires a single `SELECT` statement.

MySQL automatically deletes all temporary tables once the session is ended.

Syntax

```
CREATE TEMPORARY TABLE temp_table_name (  
    column1 datatype constraints,  
    column2 datatype constraints,  
    ...  
);
```

Temporary Tables

To create a temporary table whose structure is based on an existing table

Syntax

```
CREATE TEMPORARY TABLE temp_table_name  
SELECT statement  
...;
```

Syntax

```
DROP TEMPORARY TABLE table_name;
```

Temporary Tables

Example

```
CREATE TEMPORARY TABLE credits(  
    customerNumber INT PRIMARY KEY,  
    creditLimit DEC(10, 2)  
);  
  
INSERT INTO  
    credits(customerNumber, creditLimit)  
SELECT  
    customerNumber,    creditLimit  
FROM  
    customers WHERE    creditLimit > 0;
```

TempTable: credits

customerNumber	creditLimit
103	21000.00
112	71800.00
114	117300.00
119	118200.00
121	81700.00
124	210500.00
128	59700.00
129	64600.00

Temporary Tables

Example

```
CREATE TEMPORARY TABLE credits
SELECT
    customerNumber,
    creditLimit
FROM
    customers
WHERE
    creditLimit > 0;
```

TempTable: credits

customerNumber	creditLimit
103	21000.00
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5.5 Generated Column attribute

Columns are generated because the data in these columns are computed based on **predefined expressions (column)**.

Example

```
CREATE TABLE contacts (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    fullname varchar(101) GENERATED ALWAYS AS  
                                (CONCAT(first_name, ' ', last_name)),  
    email VARCHAR(100) NOT NULL  
);
```

insert columns programmatically

Generated Column attribute

id	first_name	last_name	fullname	email
NULL	NULL	NULL	NULL	NULL



```
INSERT INTO contacts(first_name,last_name, email)
VALUES('john','doe','john.doe@mysqлтutorial.org');
```



id	first_name	last_name	fullname	email
1	john	doe	john doe	john.doe@mysqлтutorial.org
NULL	NULL	NULL	NULL	NULL

5.6 Copy Table

Syntax: In the same database

```
CREATE TABLE new_table  
AS  
SELECT *  
FROM original_table;
```

Syntax: From another database

```
CREATE TABLE db2.new_table  
AS  
SELECT *  
FROM db1.original_table;
```